```
                                                                      ___  ____  ____  ____  ____(R)
                                                                     /__    /   ____/   /   ____/
                                                                    ___/   /   /___/   /   /___/
                                                                      Statistics/Data analysis

                                                         User: Adit Kashyap and Onofrio Cardo Log File
                                                                     Project: ECO375 Project Submission
      Source |       SS           df       MS      Number of obs   =       105
      name:  <unnamed>
       log:  C:\Users\aditk\Downloads\eco375 log 3.smcl
  log type:  smcl
 opened on:   4 Dec 2025, 17:41:56

1 . summ points gd log_broadcast log_non_broadcast log_funds wages uefa_qual

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
      points |        105    69.05714    13.92896         36        100
          gd |        105    25.40952     22.2342        -23         79
log_broadc~t |        105    4.665045    .6396005   3.033221   5.701814
log_non_br~t |        105     4.96022     .822289   3.174213   6.086775
   log_funds |        105    5.669416    .7508571   3.713572    6.87176
-------------+---------------------------------------------------------
       wages |        105    182.1506    100.3014      30.84    422.895
   uefa_qual |        105     .647619    .4155847          0          1

2 .
3 . reg points log_broadcast

      Source |       SS           df       MS      Number of obs   =       105
-------------+----------------------------------   F(1, 103)       =     10.26
       Model |  1828.11957         1  1828.11957   Prob > F        =    0.0018
    Residual |  18349.5376       103   178.15085   R-squared       =    0.0906
-------------+----------------------------------   Adj R-squared   =    0.0818
       Total |  20177.6571       104  194.015934   Root MSE        =    13.347

------------------------------------------------------------------------------
      points | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
log_broadcast |   6.555062   2.046296     3.20   0.002     2.496717    10.61341
       _cons |   38.47748   9.634522     3.99   0.000     19.36968    57.58528
------------------------------------------------------------------------------

4 .
5 . reg points log_funds

      Source |       SS           df       MS      Number of obs   =       105
-------------+----------------------------------   F(1, 103)       =     17.98
       Model |  2999.36058         1  2999.36058   Prob > F        =    0.0000
    Residual |  17178.2966       103  166.779578   R-squared       =    0.1486
-------------+----------------------------------   Adj R-squared   =    0.1404
       Total |  20177.6571       104  194.015934   Root MSE        =    12.914

------------------------------------------------------------------------------
      points | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
   log_funds |   7.152212   1.686543     4.24   0.000     3.807351    10.49707
       _cons |   28.50828   9.644417     2.96   0.004      9.38085     47.6357
------------------------------------------------------------------------------

6 .
7 . reg points wages

      Source |       SS           df       MS      Number of obs   =       105
-------------+----------------------------------   F(1, 103)       =     11.66
       Model |  2051.35549         1  2051.35549   Prob > F        =    0.0009
    Residual |  18126.3017       103  175.983511   R-squared       =    0.1017
-------------+----------------------------------   Adj R-squared   =    0.0929
       Total |  20177.6571       104  194.015934   Root MSE        =    13.266

------------------------------------------------------------------------------
      points | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       wages |   .0442789   .0129692     3.41   0.001     .0185576    .0700002
       _cons |   60.99171   2.693826    22.64   0.000     55.64915    66.33428
------------------------------------------------------------------------------
```

```
 8 .
 9 . reg points log_non_broadcast

      Source |       SS           df       MS      Number of obs   =       105
-------------+----------------------------------   F(1, 103)       =     35.25
       Model |  5144.31329         1  5144.31329   Prob > F        =    0.0000
    Residual |  15033.3439       103  145.954795   R-squared       =    0.2550
-------------+----------------------------------   Adj R-squared   =    0.2477
       Total |  20177.6571       104  194.015934   Root MSE        =    12.081

------------------------------------------------------------------------------
> ---
           points | Coefficient  Std. err.      t    P>|t|     [95% conf. interv
> al]
-------------------+----------------------------------------------------------
> ---
log_non_broadcast |   8.553081   1.440682     5.94   0.000     5.695828    11.41
> 033
            _cons |   26.63198   7.242704     3.68   0.000     12.26779    40.99
> 618
------------------------------------------------------------------------------
> ---

10 .
11 . reg points uefa_qual

      Source |       SS           df       MS      Number of obs   =       105
-------------+----------------------------------   F(1, 103)       =    168.21
       Model |  12514.5055         1  12514.5055   Prob > F        =    0.0000
    Residual |  7663.15164       103  74.3995305   R-squared       =    0.6202
-------------+----------------------------------   Adj R-squared   =    0.6165
       Total |  20177.6571       104  194.015934   Root MSE        =    8.6255

------------------------------------------------------------------------------
      points | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
   uefa_qual |   26.39555   2.035208    12.97   0.000     22.35919     30.4319
       _cons |   51.96288   1.563904    33.23   0.000     48.86125    55.06452
------------------------------------------------------------------------------

12 .
13 . xtreg points log_broadcast log_non_broadcast wages uefa_qual log_funds, fe vce
> (cluster club_id)

Fixed-effects (within) regression               Number of obs     =        105
Group variable: club_id                         Number of groups  =          6

R-squared:                                      Obs per group:
     Within  = 0.4487                                         min =         11
     Between = 0.6845                                         avg =       17.5
     Overall = 0.4867                                         max =         21

                                                F(5, 5)           =     179.35
corr(u_i, Xb) = 0.3151                          Prob > F          =     0.0000

                                    (Std. err. adjusted for 6 clusters in club_
> id)
------------------------------------------------------------------------------
> ---
                   |               Robust
           points | Coefficient  std. err.      t    P>|t|     [95% conf. interv
> al]
-------------------+----------------------------------------------------------
> ---
    log_broadcast |   7.968026   3.326018     2.40   0.062     -.581775    16.51
> 783
log_non_broadcast |   1.526276   2.867549     0.53   0.617    -5.844994    8.897
> 546
            wages |  -.0503158   .0398917    -1.26   0.263    -.1528607    .0522
> 291
        uefa_qual |   18.92041   1.749309    10.82   0.000     14.42367    23.41
> 716
        log_funds |  -5.193054   2.524863    -2.06   0.095    -11.68342    1.297
```

```
> 313
            _cons |    50.6687   17.63765     2.87   0.035     5.329685    96.00
> 771
------------------+----------------------------------------------------------------
> ---
          sigma_u |  8.2586491
          sigma_e |   7.864556
              rho |  .52442801   (fraction of variance due to u_i)
------------------------------------------------------------------------------
> ---

14 .
15 . xtreg points log_broadcast log_non_broadcast wages uefa_qual log_funds i.club_
> id, re vce(cluster club_id)

Random-effects GLS regression                   Number of obs     =        105
Group variable: club_id                         Number of groups  =          6

R-squared:                                      Obs per group:
     Within  = 0.4487                                         min =         11
     Between = 1.0000                                         avg =       17.5
     Overall = 0.7119                                         max =         21

                                                Wald chi2(5)      =          .
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =          .

                                    (Std. err. adjusted for 6 clusters in clu
> b_id)
------------------------------------------------------------------------------
> ------
                 |               Robust
              points | Coefficient  std. err.      z    P>|z|     [95% conf. inte
> rval]
------------------+-----------------------------------------------------------
> ------
      log_broadcast |   7.968026    3.41333     2.33   0.020     1.278023    14.
> 65803
  log_non_broadcast |   1.526276   2.942826     0.52   0.604    -4.241557    7.2
> 94108
              wages |  -.0503158   .0409389    -1.23   0.219    -.1305546     .0
> 29923
          uefa_qual |   18.92041    1.79523    10.54   0.000     15.40183      2
> 2.439
          log_funds |  -5.193054   2.591144    -2.00   0.045     -10.2716   -.11
> 45059
                    |
            club_id |
            Chelsea |    4.40679   2.230432     1.98   0.048     .0352234    8.7
> 78357
            Everton |  -9.432999    4.57089    -2.06   0.039    -18.39178   -.47
> 42188
   Manchester City  |   15.49935   3.200402     4.84   0.000     9.226678    21.
> 77202
     Manchester UN  |   1.957528   2.898754     0.68   0.499    -3.723924    7.6
> 38981
Tottenham Hotspurs  |  -2.347212   1.251761    -1.88   0.061    -4.800617    .10
> 61939
                    |
              _cons |    50.2969   19.20788     2.62   0.009     12.65015    87.
> 94364
------------------+-----------------------------------------------------------
> ------
            sigma_u |          0
            sigma_e |   7.864556
                rho |          0   (fraction of variance due to u_i)
------------------------------------------------------------------------------
> ------
```

```
16 .
17 . xtreg points log_broadcast log_non_broadcast wages uefa_qual log_funds i.club_
   > id, re vce(cluster club_id)

   Random-effects GLS regression                   Number of obs     =        105
   Group variable: club_id                         Number of groups  =          6

   R-squared:                                      Obs per group:
        Within  = 0.4487                                         min =         11
        Between = 1.0000                                         avg =       17.5
        Overall = 0.7119                                         max =         21

                                                   Wald chi2(5)      =          .
   corr(u_i, X) = 0 (assumed)                      Prob > chi2       =          .

                                       (Std. err. adjusted for 6 clusters in clu
   > b_id)
   ------------------------------------------------------------------------------
   > ------
                        |               Robust
                 points | Coefficient  std. err.      z    P>|z|     [95% conf. inte
   > rval]
   ------------------------------------------------------------------------------
   > ------
          log_broadcast |   7.968026    3.41333     2.33   0.020     1.278023    14.
   > 65803
      log_non_broadcast |   1.526276   2.942826     0.52   0.604    -4.241557    7.2
   > 94108
                  wages | -.0503158   .0409389    -1.23   0.219    -.1305546     .0
   > 29923
              uefa_qual |   18.92041    1.79523    10.54   0.000     15.40183      2
   > 2.439
              log_funds | -5.193054   2.591144    -2.00   0.045     -10.2716   -.11
   > 45059
                        |
                 club_id |
                Chelsea |    4.40679   2.230432     1.98   0.048     .0352234    8.7
   > 78357
                Everton | -9.432999    4.57089    -2.06   0.039    -18.39178   -.47
   > 42188
       Manchester City  |   15.49935   3.200402     4.84   0.000     9.226678    21.
   > 77202
          Manchester UN |   1.957528   2.898754     0.68   0.499    -3.723924    7.6
   > 38981
   Tottenham Hotspurs   | -2.347212   1.251761    -1.88   0.061    -4.800617    .10
   > 61939
                        |
                   _cons |    50.2969   19.20788     2.62   0.009     12.65015    87.
   > 94364
   ------------------------------------------------------------------------------
   > ------
                sigma_u |          0
                sigma_e |   7.864556
                    rho |          0   (fraction of variance due to u_i)
   ------------------------------------------------------------------------------
   > ------

18 .
19 . margins, at(log_broadcast = (10(1)15))

   Predictive margins                                      Number of obs = 105
   Model VCE: Robust

   Expression: Linear prediction, predict()
   1._at: log_broadcast = 10
   2._at: log_broadcast = 11
   3._at: log_broadcast = 12
   4._at: log_broadcast = 13
   5._at: log_broadcast = 14
   6._at: log_broadcast = 15
```

| | Margin | Delta-method std. err. | z | P>\|z\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| _at | | | | | | |
| 1 | **111.5662** | **18.20996** | **6.13** | **0.000** | **75.87534** | **147.2571** |
| 2 | **119.5342** | **21.62329** | **5.53** | **0.000** | **77.15336** | **161.9151** |
| 3 | **127.5023** | **25.03662** | **5.09** | **0.000** | **78.43138** | **176.5731** |
| 4 | **135.4703** | **28.44995** | **4.76** | **0.000** | **79.7094** | **191.2312** |
| 5 | **143.4383** | **31.86328** | **4.50** | **0.000** | **80.98743** | **205.8892** |
| 6 | **151.4063** | **35.27661** | **4.29** | **0.000** | **82.26545** | **220.5472** |

```
20 .
21 . marginsplot, xdimension(log_broadcast) ytitle("Predicted points") xtitle("log(
> broadcasting revenue)") title("Predicted points vs broadcast revenue")
```

Variables that uniquely identify margins: **log_broadcast**

```
22 .
23 . margins club_id
```

Predictive margins  Number of obs = **105**
Model VCE: **Robust**

Expression: **Linear prediction, predict()**

```
> ──────
```

| | Margin | Delta-method std. err. | z | P>\|z\| | [95% conf. | inte |
|---|---|---|---|---|---|---|
| club_id | | | | | | |
| Arsenal | **68.68534** | **1.20899** | **56.81** | **0.000** | **66.31576** | **71.** |
| Chelsea | **73.09213** | **1.422275** | **51.39** | **0.000** | **70.30452** | **75.** |
| Everton | **59.25234** | **3.811412** | **15.55** | **0.000** | **51.78211** | **66.** |
| Manchester City | **84.18469** | **2.909055** | **28.94** | **0.000** | **78.48305** | **89.** |
| Manchester UN | **70.64287** | **2.651421** | **26.64** | **0.000** | **65.44618** | **75.** |
| Tottenham Hotspurs | **66.33813** | **1.984757** | **33.42** | **0.000** | **62.44808** | **70.** |

```
> ──────
24 .
25 . marginsplot, xdimension(club_id) ytitle("Predicted points") xtitle("Club") tit
> le("Predicted points by club (RE model)")
```

Variables that uniquely identify margins: **club_id**

```
26 .
27 . twoway (scatter points log_broadcast) (lfit points log_broadcast), title("Poin
> ts and Log Broadcast Revenue with Fitted Line") xtitle("Log Broadcast Revenue"
> ) ytitle("League Points")

28 .
29 . twoway scatter points log_broadcast, by(club, title("Points vs Log Broadcast R
> evenue by Club")) xtitle("Log Broadcast Revenue") ytitle("League Points")

30 . log close
```

name: **<unnamed>**
log: **C:\Users\aditk\Downloads\eco375 log 3.smcl**
log type: **smcl**
closed on: **4 Dec 2025, 17:42:55**